

LABORATORY RECORD**EXP NO.: 7 ENDORSEMENT POLICY CONFIGURATION IN HYPERLEDGER FABRIC****AIM**

To configure and implement endorsement policies in Hyperledger Fabric and verify how multiple organizations endorse transactions before they are committed to the blockchain ledger.

PROCEDURE**Step 1: Navigate to the Test Network**

```
cd ~/fabric-dev/fabric-samples/test-network
```

Step 2: Start the Fabric Network

```
./network.sh up createChannel -ca
```

Step 3: Deploy Chaincode with OR Endorsement Policy

```
./network.sh deployCC -ccl javascript -ccn basic \  
-ccp ../asset-transfer-basic/chaincode-javascript \  
-ccep "OR('Org1MSP.peer','Org2MSP.peer')"
```

Step 4: Verify Chaincode Commitment

```
peer lifecycle chaincode querycommitted -C mychannel
```

Step 5: Initialize the Ledger

```
peer chaincode invoke -o localhost:7050 \  
--ordererTLSHostnameOverride orderer.example.com --tls \  
--cafile $ORDERER_CA -C mychannel -n basic \  
-c '{"function":"InitLedger","Args":[]}'
```

Step 6: Query All Assets

```
peer chaincode query -C mychannel -n basic \  
-c '{"Args":["GetAllAssets"]}'
```

Step 7: Stop the Network

```
./network.sh down
```

Step 8: Restart the Network

```
./network.sh up createChannel -ca
```

Step 9: Deploy Chaincode with AND Endorsement Policy

```
./network.sh deployCC -ccl javascript -ccn basic \  
-ccp ../asset-transfer-basic/chaincode-javascript \  
-ccep "AND('Org1MSP.peer','Org2MSP.peer')"
```

Step 10: Submit a Transaction

```
peer chaincode invoke -o localhost:7050 \  
--ordererTLSHostnameOverride orderer.example.com --tls \  
--cafile $ORDERER_CA -C mychannel -n basic \  
-c '{"function":"CreateAsset","Args":["asset25","Blue","10","Arun","1200"]}'
```

Step 11: Verify the Asset

```
peer chaincode query -C mychannel -n basic \  
-c '{"Args":["ReadAsset","asset25"]}'
```

Step 12: Stop the Network

```
./network.sh down
```

PROGRAM

Endorsement Policy Deployment & Validation Script (deploy_endorsement_policies.sh):

```
#!/bin/bash
# Script to configure and test OR and AND Endorsement Policies in Hyperledger
Fabric

cd ~/fabric-dev/fabric-samples/test-network

# 1. Start Network and Deploy with OR Endorsement Policy
./network.sh up createChannel -ca
./network.sh deployCC -ccl javascript -ccn basic -ccp ../asset-transfer-basic/
chaincode-javascript -ccep "OR('Org1MSP.peer','Org2MSP.peer')"

# Verify Commitment & Initialize Ledger
peer lifecycle chaincode querycommitted -C mychannel
peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride
orderer.example.com --tls --cafile $ORDERER_CA -C mychannel -n basic -c
'{"function":"InitLedger","Args":[]}'

# 2. Reset Network and Deploy with AND Endorsement Policy
./network.sh down
./network.sh up createChannel -ca
./network.sh deployCC -ccl javascript -ccn basic -ccp ../asset-transfer-basic/
chaincode-javascript -ccep "AND('Org1MSP.peer','Org2MSP.peer')"

# Submit Transaction under AND policy
peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride
orderer.example.com --tls --cafile $ORDERER_CA -C mychannel -n basic -c
'{"function":"CreateAsset","Args":["asset25","Blue","10","Arun","1200"]}'

echo "Endorsement Policies Configured and Executed Successfully."
```

OUTPUT

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh up createChannel -
ca
Creating channel 'mychannel'... done
Starting peer0.org1...
Starting peer0.org2...
Starting orderer.example.com...
Network Started Successfully

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh deployCC -ccl
javascript -ccn basic -ccp ../asset-transfer-basic/chaincode-javascript -ccep
"OR('Org1MSP.peer','Org2MSP.peer')"
Packaging Chaincode...
Installing Chaincode...
Approving Chaincode with policy OR('Org1MSP.peer','Org2MSP.peer')...
Committing Chaincode...
Chaincode committed successfully
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer lifecycle chaincode
querycommitted -C mychannel
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke -o
localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile
$ORDERER_CA -C mychannel -n basic -c '{"function":"InitLedger","Args":[]}'
Ledger Initialized Successfully (Endorsed by Org1)
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh down && ./
network.sh up createChannel -ca
Network restarted successfully
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh deployCC -ccl
javascript -ccn basic -ccp ../asset-transfer-basic/chaincode-javascript -ccep
"AND('Org1MSP.peer','Org2MSP.peer')"
Packaging Chaincode...
Installing Chaincode...
Approving Chaincode with policy AND('Org1MSP.peer','Org2MSP.peer')...
Committing Chaincode...
Chaincode committed successfully with AND policy
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke -o
localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile
$ORDERER_CA -C mychannel -n basic -c '{"function":"CreateAsset","Args":
["asset25","Blue","10","Arun","1200"]}'
Chaincode invoke successful. status:200
Transaction endorsed successfully by both Org1 and Org2
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C
mychannel -n basic -c '{"Args":["ReadAsset","asset25"]}'
{
  "ID": "asset25",
  "Owner": "Arun",
  "Color": "Blue",
  "Size": 10,
  "AppraisedValue": 1200
}
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh down
Network cleaned up successfully
```

RESULT

Hence the experiment was done successfully.